

CSC411 - Software Engineering Test  
Answer all Questions      Time: 30 mins

- Q1 Suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: 3mks
- i. A system to control antilock braking in a car
  - ii. A university accounting system that replaces an existing system
- Q2 Mention four (4) reasons why you model a process. 2mks
- Q3. Describe the following types of Process models (remember to include when to use, diagram advantages and disadvantages) 5mks
- i. Iterative Process Model 5mks
  - ii. Incremental Process Model



**CSC411 (SOFTWARE ENGINEERING) TEST**  
**INSTRUCTION: ANSWER ALL QUESTIONS**      **TIME: 40MINS**

**Section A (Requirements Engineering - Objectives)    10mks**

**1. Which one of the following is a functional requirement?**

- a) Maintainability    b) Portability    c) Robustness    d) None of the options    e) All the options

**2. Which one of the following is a requirement that fits in a developer's module ?**

- a) Availability    b) Testability    c) Usability    d) Flexibility

**3. "Consider a system where, a heat sensor detects an intrusion and alerts the security company." What kind of a requirement the system is providing?**

- a) Functional    b) Non-Functional    c) Known Requirement    d) None of the options

**4. Which of the following statements explains portability in non-functional requirements?**

- a) It is a degree to which software running on one platform can easily be converted to run on another platform  
b) It cannot be enhanced by using languages, OS' and tools that are universally available and standardized  
c) The ability of the system to behave consistently in a user-acceptable manner when operating within the environment for which the system was intended  
d) None of the options

**5. What are the four dimensions of Dependability?**

- a) Usability, Reliability, Security, Flexibility    b) Availability, Reliability, Maintainability, Security  
c) Availability, Reliability, Security, Safety    d) Security, Safety, Testability, Usability

**6. What is the first step of requirement elicitation?**

- a) Identifying Stakeholder    b) Listing out Requirements    c) Requirements Gathering  
d) All of the options

**7. Why is Requirements Elicitation a difficult task?**

- a) Problem of scope    b) Problem of understanding    c) Problem of volatility  
d) All of the options

**8. The process to gather the software requirements from client, analyze and document them is known as \_\_\_\_.**

- a) Feasibility Study    b) Requirement Gathering    c) Requirement Engineering  
d) System Requirements Specification

**9. The System Requirement Specification (SRS) is said to be \_\_\_\_\_ if and only if no subset of individual requirement described in it conflict with each other**

- a) Correct    b) Consistent    c) Unambiguous    d) Verifiable

**10. If every requirement can be checked by a cost-effective process, then SRS is \_\_\_\_**

- a) Traceable    b) Verifiable    c) Modifiable    d) Complete    e) Correct



## Section B (System Modelling - UML Diagrams)

### **1. Use case Diagrams for ATM**

An automated teller machine (ATM) is a banking subsystem (**subject**) that provides bank customers with access to financial transactions in a public space without the need for a cashier, clerk, or bank teller.

*A Customer* uses bank ATM to *Check Balances* of his/her bank accounts, *Deposit Funds*, *Withdraw Cash* and/or *Transfer Funds*.

*ATM Technician* provides *Maintenance* and *Repairs*.

All these use cases also involve **Bank actor** whether it is related to customer transactions or to the ATM servicing.

**Provide:**

- i. A well-drawn use diagram for top-level uses cases given above 4mks
- ii. A separate use case diagram for **Bank ATM Maintenance, Repair, Diagnostics** given the following additional information 3mks

ATM Technician maintains or repairs Bank ATM. Maintenance use case includes Replenishing ATM with cash, ink or printer paper, Upgrades of hardware, firmware or software, and remote or on-site Diagnostics. Diagnostics is also included in Repair use case.

### **2. Class Diagram for Course-Information-System**

The lecturers and courses details are listed in a prospectus in which courses or lecturers can be added or removed. Courses at the college comprise a number of units at two levels, with students normally taking level one units in the first year of their course and progressing level two units in the second. Part time students may take two or more years to complete all their units at a given level. Level one units are assessed on course work grades, but level two units are assessed with examinations. Some courses also include a compulsory foundation unit where particular technical skills need to be established.

The system will provide lists of lecturers, courses and units, and access to more detailed information about which units are taught on which courses. Individual lecturer records may be queried for information such as their office room number, e-mail address and subject specialisms, as well as reporting the units that a lecturer teaches. Each course in the prospectus can be identified by a unique course code, and consists of a number of units at each level. Every course has one lecturer acting as course leader, who is responsible for administering that course and updating unit descriptions. Management functions of the prospectus allow lecturers and courses to be added to or removed from the system."

- i. Extract the main objects (entities) for above system. 1mk
- ii. Find the necessary attributes and functions that need to be associated with each object (mention at most three attributes and one function for a class). 2mks
- iii. Identify the relationships between these objects. 1mk
- iv. **Construct a final comprehensive Class diagram showing all objects and their relationships along with their attributes and functions.** 4mks





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**B.Sc DEGREE EXAMINATION**

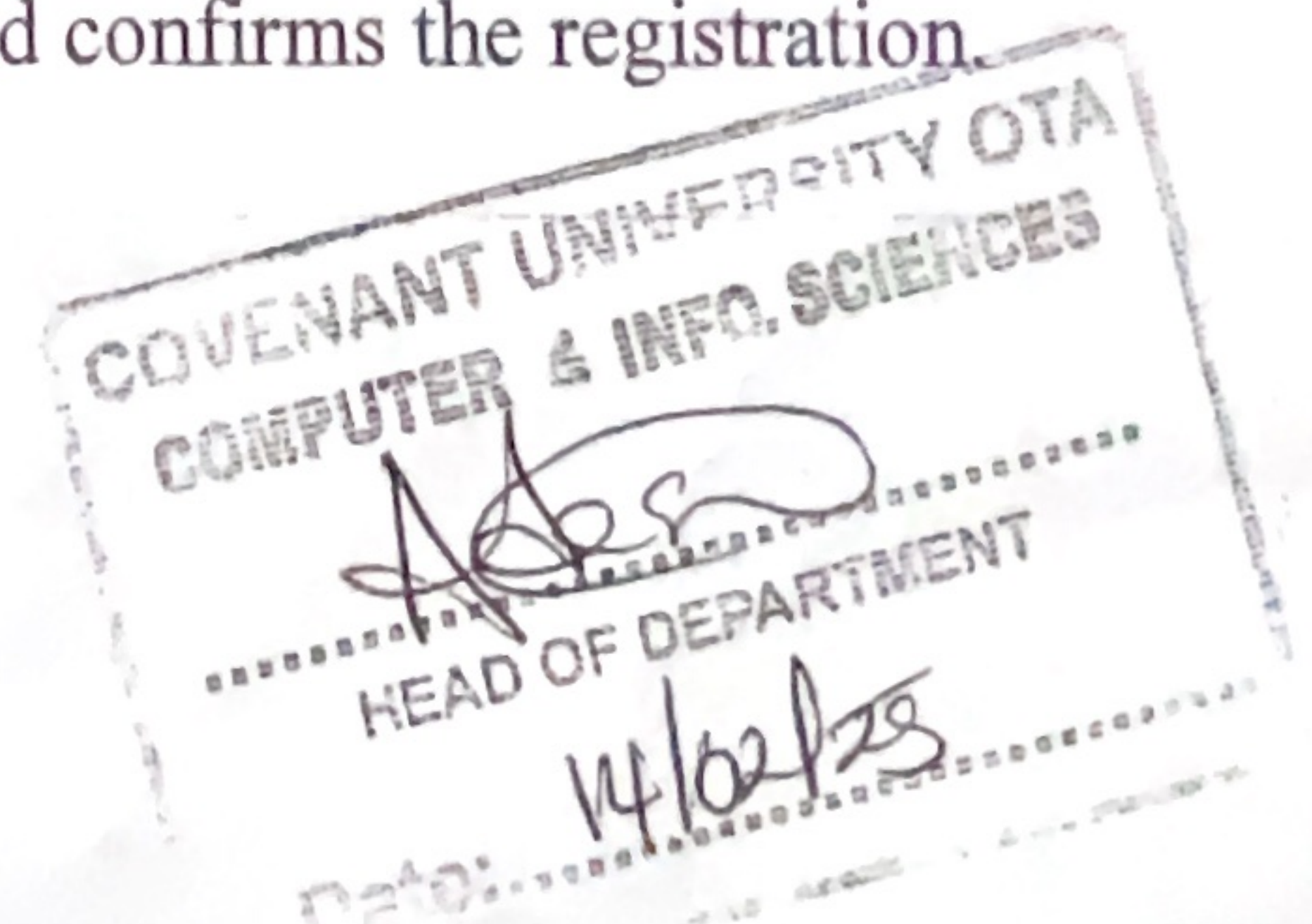
**COLLEGE: SCIENCE AND TECHNOLOGY**    **DEPARTMENT: Computer & Info Sciences**  
**SESSION: 2024/2025**    **SEMESTER: ALPHA**  
**COURSE CODE: CSC411**    **CREDIT UNIT: 3**  
**COURSE TITLE: SOFTWARE ENGINEERING**    **TIME: 3 HRS**  
**INSTRUCTION: ANSWER ANY FOUR QUESTIONS**

**Question One**

- a) Describe when to use each of the following: **Waterfall model, Incremental SDLC Model, and Spiral SDLC Model** **6mks**
- b) Assume student registration system with classes: Student, Session, Registrar, Transcript, Department, Course and CourseGrade. Draw a class diagram with the following relationship: (Provide the attributes and methods for each class in your diagram) **6mks**
- A student can only collect one transcript  
All students can only registrar in one registrar's office  
All records for the session can only be found in registrar's office  
All students belong to a Department  
All Courses belong to between 1 and 3 Departments.  
Each course belongs to a CourseGrade
- c) i) Given that the functional requirements of a student admission system are as follows: Apply for admission, Enroll in the university, Enroll in a course, Withdraw from a course, draw the use case diagram. **2½ mks**
- ii) Draw a sequence diagram for registering a course by student. **3mks**

**Question Two**

- a) Explain how an Agile software process differs from a Traditional software process and give one advantage and one disadvantage of each. **4mks**
- b) i) Identify the **actors, objects** and at **least two possible assumptions** in the following scenario to register a patient in a hospital management system: **5½mks**
- The administrator enters the patient's name, address, date of birth and emergency contact details into the system. If the patient has only public health insurance, the administrator enters the patient's medicare number, and the system verifies this with government health database. If the patient also has private health insurance, then the administrator enters also the patient's private health insurance details, and the system verifies these details with the private health insurance system. When these details are verified as correct, the system saves the patient's details and confirms the registration.





- ii) Refer to Question 2b(i) and sketch a UML sequence diagram for the scenario where the administrator registers a patient who only has public health insurance. 4mks
- c) i) Differentiate between the following System Modeling and Requirements Engineering 2mks  
 ii) Give any two (2) reasons why eliciting requirements is difficult 1mk  
 iii) Provide any two (2) items that should be included in a scenario 1mk

### Question Three

- a) i) Give any three (3) reasons why requirements negotiation is needed in software engineering. 1½mks
- ii) A security light system has a switch and a motion sensor attached. It can be either armed or unarmed. If the switch is in the off position the light is off and the system is unarmed. When the switch is turned on, the light stays off but the system is armed. If the system is armed and the motion sensor detects movement, the light comes on. If no movement is detected for 5 seconds, the light goes off. Draw a UML State-chart diagram to describe the states of the security light system. 4½mks
- b) Consider the information below.  
 Pair programming is an agile software development technique in which two programmers work together at one work station. One types in code while the other reviews each line of code as it is typed in. The person typing is called the **driver**. The person reviewing the code is called the **observer**. The two programmers switch roles frequently (possibly every 30 minutes or less).  
 Suppose that you are asked to build a system that allows **Remote Pair Programming**. That is, the system should allow the driver and the observer to be in remote locations, but both can view a single desktop in real-time. The driver should be able to edit code and the observer should be able to "point" to objects on the driver's desktop. In addition, there should be a video chat facility to allow the programmers to communicate. The system should allow the programmers to easily swap roles and record rationale in the form of video chats. In addition, the driver should be able to issue the system to backup old work.
- i) Draw a use case diagram to show all the functionalities of the system 3mks
- ii) Provide three non-functional requirements of the system 3mks  
 internet speed is slow and when the internet connection is suddenly down the user should be able to resume the session at a later time.
- c) i) How is the practice of software engineering different from the practice of software development? 2½mks
- ii) Given the integration and configuration process model in Figure 1, explain why it is necessary to repeat the requirements engineering activity in the process. Hence describe the two activities involved in this process. 3mks

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14/02/25



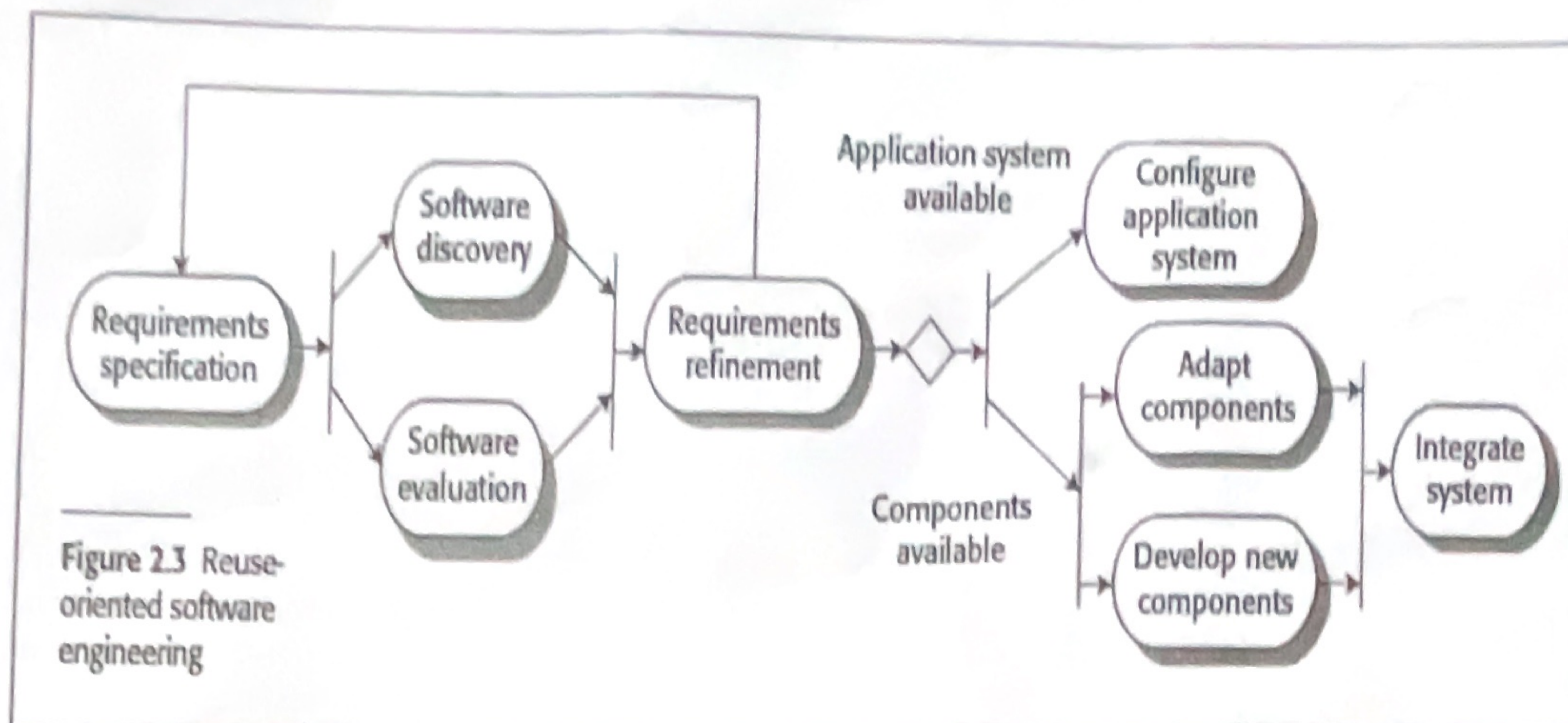


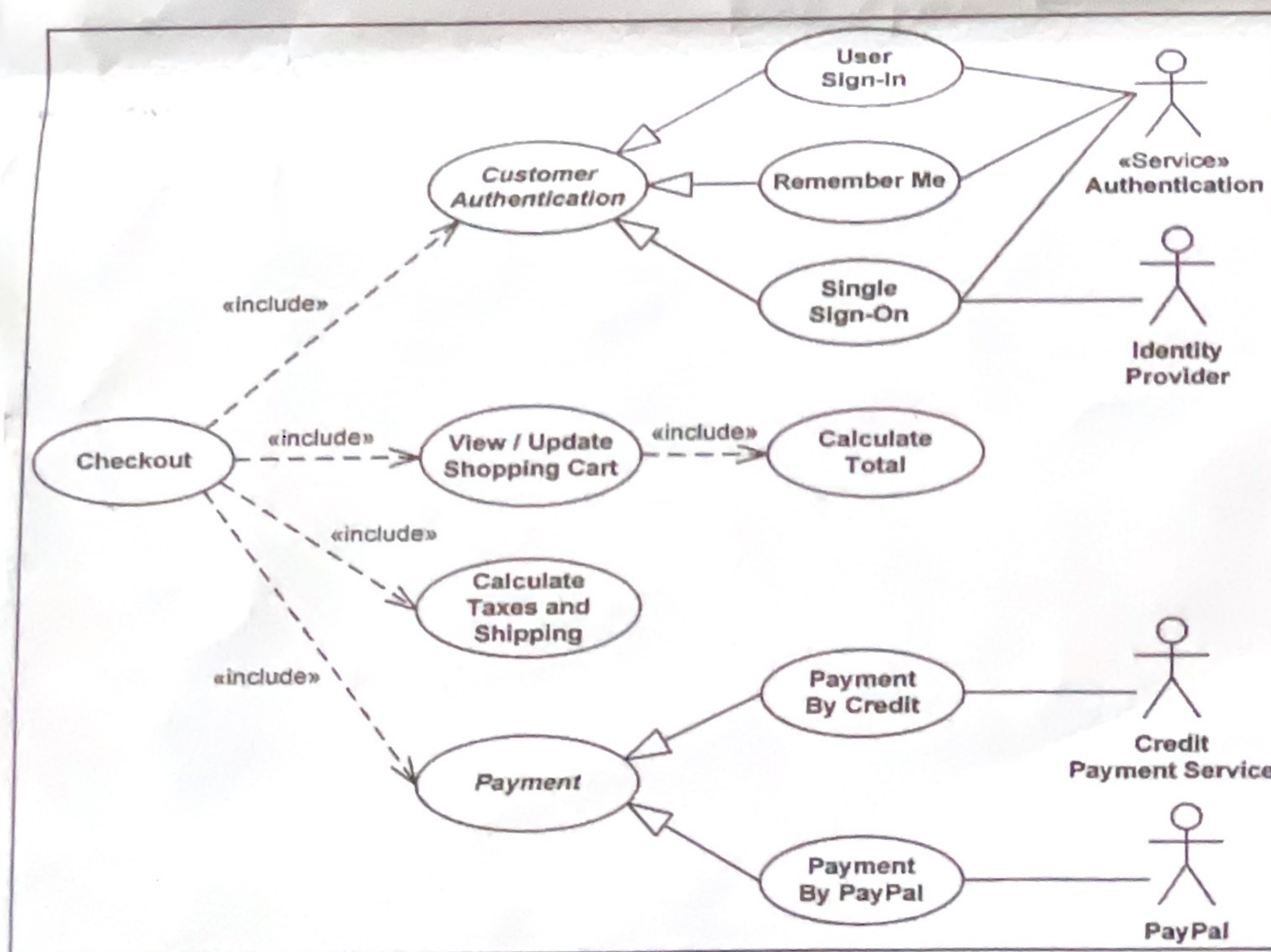
Figure 1

#### Question Four

- a) i) Provide a **top-level** use case diagram for an **Online Shopping system** given the following statements **6mks**

A Web Customer actor can be a New user or Registered User who uses some web site to make purchases online. Only registered user is able to View Items and Make Purchase. All New users must undergo Client registration on the website. View Items use case could be used by customer as top-level use case if customer only wants to find and see some products however, View items use case must be a part of Make Purchase use case just as the Checkout use case is also a part of the Make purchase use case. Apart from the Web Customer actor there are several other actors like Service Authentication, Identity Provider, and Credit payment service. The check out and view item actions must be authenticated by the Service authentication and Identity provider.

- b) Provide detail description of the following Checkout use case diagram **4mks**





- c) i) Software engineering is an engineering discipline that is concerned with all aspects of software production. Briefly discuss the fundamental activities that take place in software engineering. **2mks**
- ii) With the aid of a diagram, describe a client – server architecture for a film library **3mks.**
- iii) Mention any five ethical behaviors that should be upheld by software engineers. **2½mks**

#### Question Five

- a) i) List and explain any three essential attributes of a good software **3mks**
- ii) Explain why it is important to differentiate between user requirements and system requirements in the requirements engineering process. **3mks**
- b) i) Provide a Context Diagram for a Hotel Reservation System (HRS). A brief description is given below: **4mks**
- The HRS is a computerized system that distributes and stores the hotel information. It assists hotel managers in running their sales and online marketing activities, allowing them to update their room rates and vacant rooms and making them visible on their sales channels. These comprise both online and conventional travel agencies.
- ii) Provide a Sequence diagram based on the following sequence of events: **5mks**
- A customer places an order from GreenLand Restaurant for which he/she pays to a cash counter. The cash counter is required to confirm the order after the payment. The cash counter informs the food counter to prepare the order. Finally, the food counter sends out the order delivery. **Note:** This is only the **order system** of the GreenLand.
- c) In a tabular form, state five differences between Agile Requirements Engineering and Traditional Requirements Engineering **2½mks**

#### Question Six

- a) i) Suggest and provide reasons for the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: **3mks**
1. A system to control anti-lock braking in a car
  2. A university accounting system that replaces an existing system
- ii) What are the most important differences between generic software product Development and custom software development? What might this mean in practice for users of generic software products? **4mks**
- b) i) Provide three reasons for which an organization decide to scrap a system when the system assessment suggests that it is of high quality and of high business value **3mks**
- ii) Discuss the role of requirement validation and verification in software engineering. **2mks**
- c) A company consists of departments. Departments are located in one or more offices. One office acts as a headquarter. Each department has a manager who is recruited from the set of employees. Draw a class diagram which consists of all the classes in your system their attributes and operations, relationships between the classes, multiplicity specifications, and other model elements that you find appropriate. **5½mks**